

RetailPulse - Customer Behavior Analytics & Revenue Intelligence Platform

Executive Business Intelligence Report

Source: online_retail_II.xlsx | Generated: 2026-09-29T16:04:25.182353+00:00

Executive Summary

The analysis covers 1,067,371 transaction rows. After exact-duplicate removal and the documented positive-sales rules, 779,425 completed positive-sales rows were retained for the primary customer and revenue analyses. The primary revenue measure is Quantity multiplied by Price for positive completed sales.

Metric	Value
Raw transactions	1,067,371
Cleaned sales lines	779,425
Customers	5,878
Orders	36,969
Revenue	£17,374,804
AOV	£470
Repeat customer rate	72.4%

1. Data Quality and Cleaning

The raw audit found 34,335 exact duplicate rows, 243,007 missing Customer ID values, 4,382 missing descriptions, 22,950 negative-quantity rows, 6,202 zero-price rows, 5 negative-price rows, and 19,494 cancellation-style invoice rows. Customer-level analyses exclude missing customer identifiers because lifecycle tracking is otherwise impossible.

Imputation: missing descriptions are filled with 'Unknown Product Description' (0 primary-sales rows). Customer IDs are deliberately not imputed. Robustness: IQR 1.5x helper caps are applied to Quantity and Price for analytical/modeling use, with 51,119 quantity values and 65,481 price values capped; original fields remain intact for headline revenue. Quantity cap: 1 to 27.0; Price cap: 0.01 to 7.5.

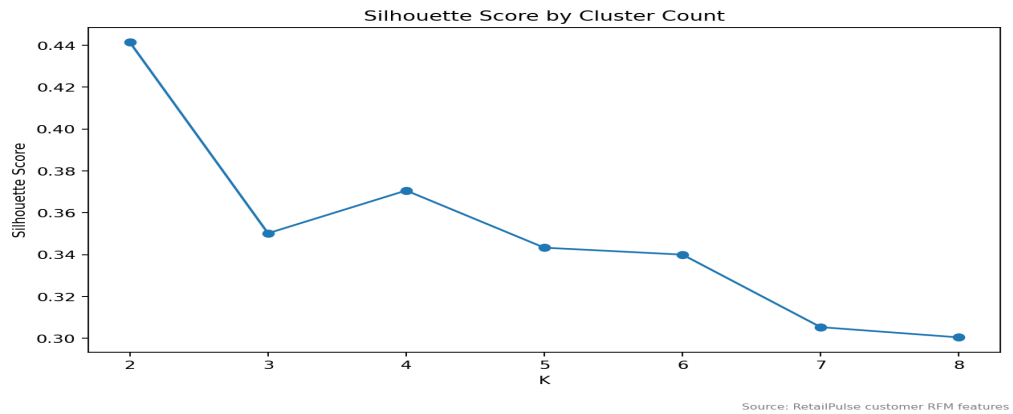
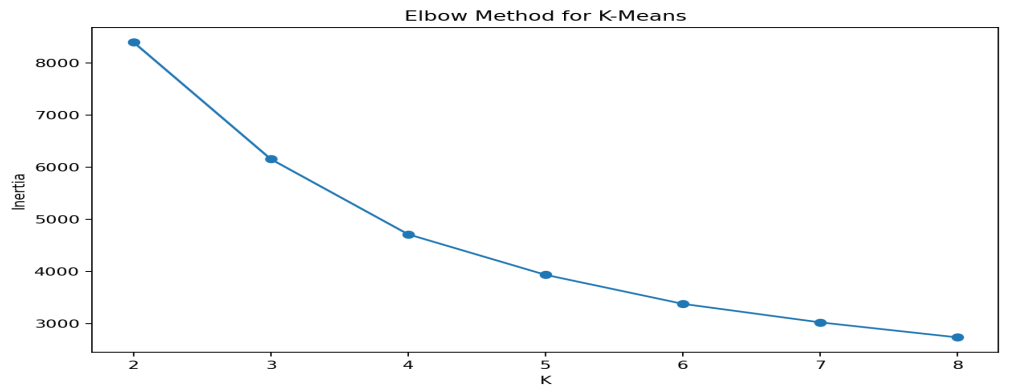
2. Exploratory Data Analysis

EDA generated 12 meaningful visualizations covering revenue, orders, quantity, price, countries, products, customer monetary value, cancellations, missingness and correlation. The top revenue country was United Kingdom with £14,389,235 of positive-sales revenue.

3. Customer Segmentation

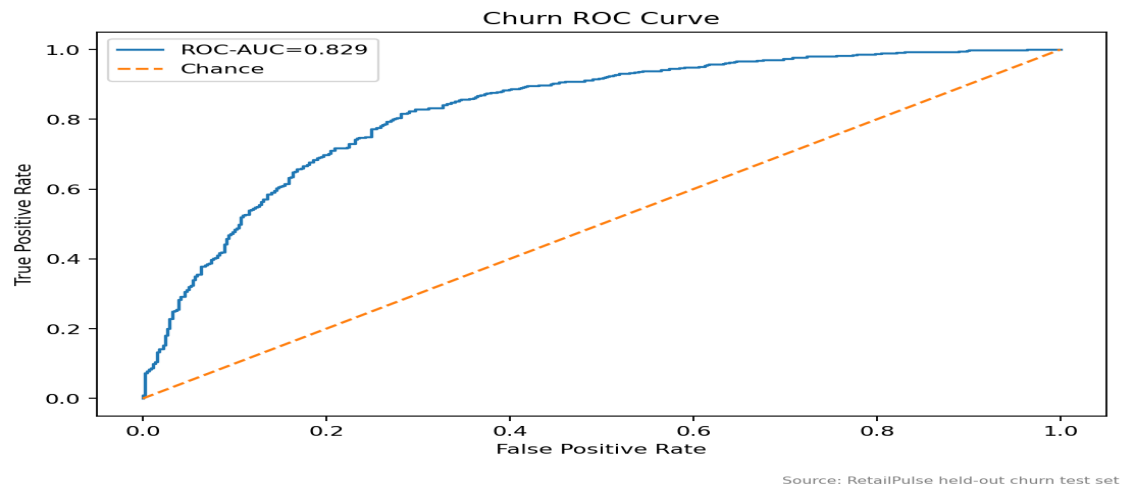
K=4 selected because the inertia reductions begin flattening after K=4 while the silhouette score remains positive; the global silhouette maximum is K=2, which is noted as a methodological limitation. The four-cluster solution is used because the official brief requires four business segment profiles. This is a documented, non-arbitrary trade-off rather than a hidden hardcoded.

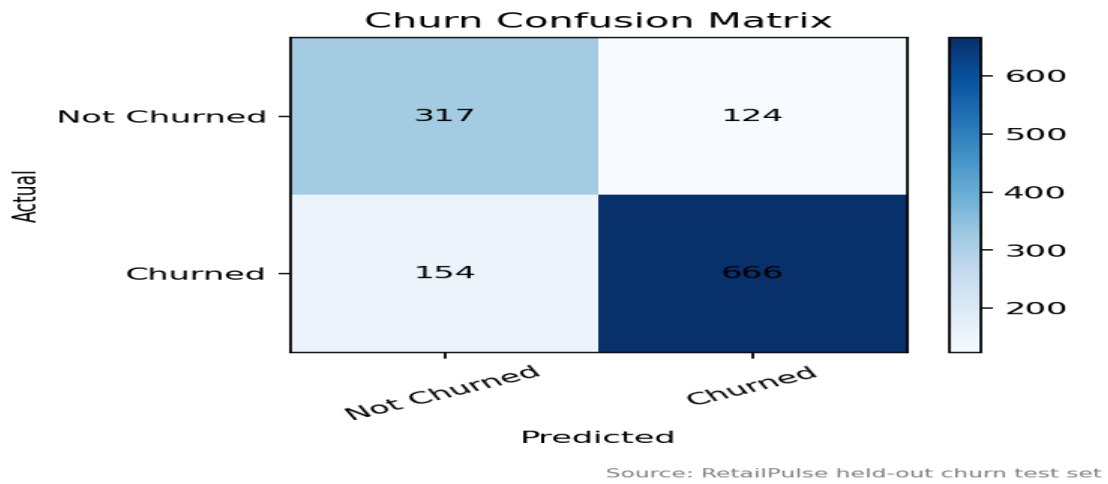
Cluster	Profile	Customers	%	Avg Recency	Avg Frequency	Avg Monetary	Avg AOV
0	Potential / Recent Buyers	1,234	21.0%	28.9	3.0	£823	£313
1	Champions & High-Value Customers	1,224	20.8%	27.9	19.0	£10,545	£562
2	At-Risk & Hibernating Customers	1,967	33.5%	396.3	1.4	£318	£250
3	Established / Lapsed Customers	1,453	24.7%	230.0	5.0	£1,946	£481



4. Churn Prediction

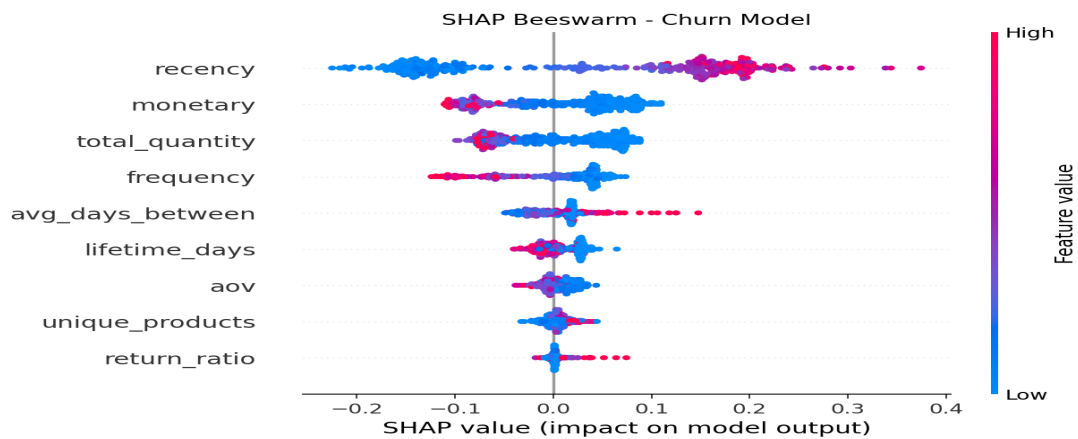
A temporal churn target was created using an observation window ending 2011-06-30 and a three-month prediction window ending 2011-09-30. The Random Forest held-out test ROC-AUC was 0.829; cross-validation mean AUC was 0.826. Precision was 0.843, recall 0.812, F1 0.827. The >0.80 target was achieved on the held-out set.





5. SHAP Explainability

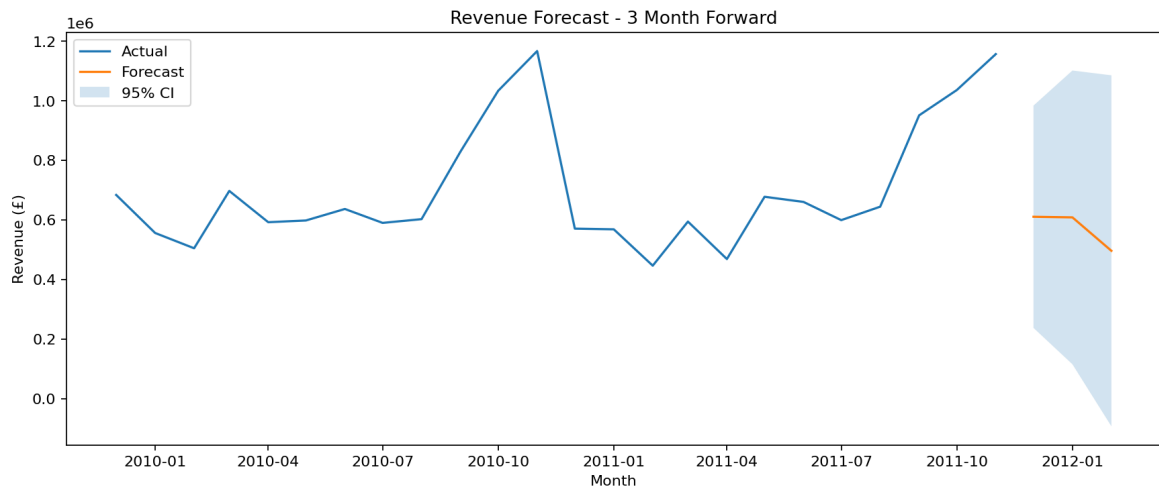
SHAP summaries explain which features most influence churn predictions. The results are predictive associations, not causal effects. The strongest mean-absolute SHAP feature is shown in the dashboard and saved under artifacts/shap.



The SHAP waterfall for a representative high-risk test customer is retained under artifacts/shap/shap_waterfall_high_risk.png and is also shown in the Streamlit dashboard.

6. Revenue Forecasting

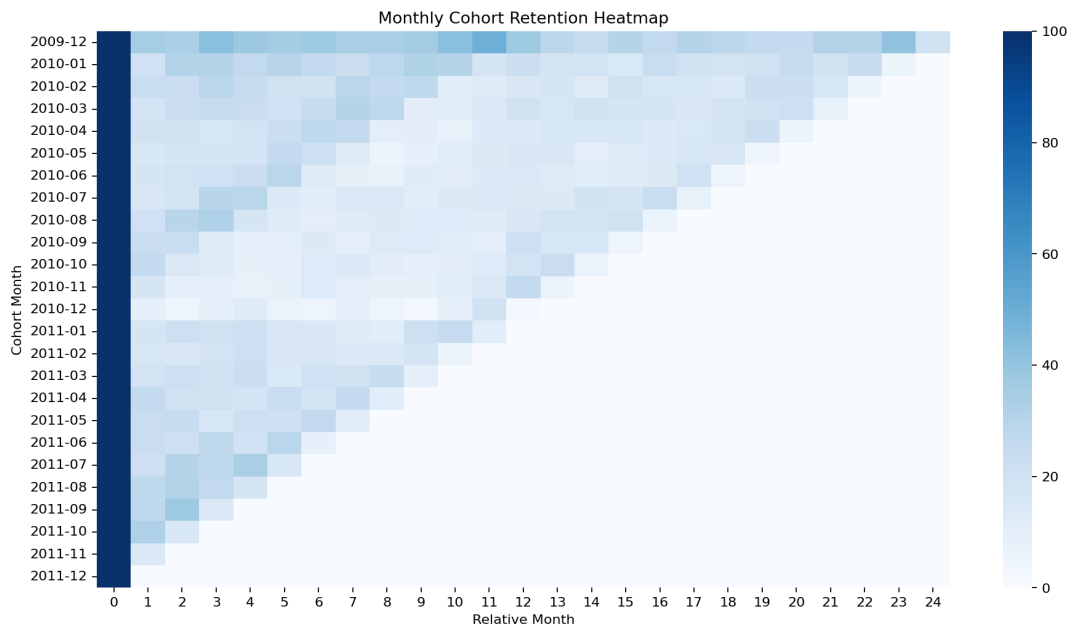
A SARIMA model was selected using a three-month historical backtest. Selected model: SARIMA(0, 1, 1)x(1, 0, 0, 12). Backtest MAPE was 3.86%, RMSE was £60,305. The final model was fitted through the last complete observed month and generated a three-month forward forecast with 95% confidence intervals. Forecast values are £610,300 for the first month and £1,714,489 cumulatively across the three-month horizon.



Source: RetailPulse monthly revenue series

7. Cohort Retention

Monthly cohorts are defined from the customer's first purchase month. Retention is the share of the original cohort that is active in each relative month. The full retention matrix is saved in `artifacts/tables/cohort_retention.csv`.



Source: RetailPulse completed purchase data

8. Statistical Testing

Test	Statistic	p-value	Significant?
Welch t-test: UK vs International AOV	-7.55	1.16e-13	Yes
Chi-square: Segment vs churn status	1.93e+03	0	Yes
One-way ANOVA: AOV across segments	21.5	8.05e-14	Yes

For every test, the null and alternative hypotheses, test statistic and p-value are also retained in `artifacts/metrics/statistical_tests.json`. Statistical significance is not treated as causation.

9. Business Recommendations

- Use RFM segments to differentiate retention, development and reactivation campaigns rather than applying one message to all customers.
- Prioritize churn interventions around customers whose recency has moved materially beyond their normal purchase cadence.
- Use SHAP explanations with model-risk safeguards so retention teams can see the behavioral features behind an individual prediction.
- Use the revenue forecast as a planning input for inventory, staffing and cash-flow preparation, while monitoring actual-versus-forecast error each month.
- Track cohort retention after acquisition campaigns so marketing teams can compare customer quality beyond first-order revenue.

10. Reproducibility and Runtime Notes

The repository contains the seven notebooks plus reusable source modules and a single `run_pipeline.py` entry point. After installing requirements, the pipeline regenerates the numerical artifacts and the report. MLflow logging is implemented in `src/mlflow_tracking.py` and requires the MLflow package at runtime. Streamlit dashboard code is implemented in `dashboard/app.py` and requires Streamlit at runtime. These are environment/account actions and are not silently represented as completed when the package is unavailable in the build environment.

11. Limitations

The churn label is operational rather than a directly observed business churn field. The dataset's final month is incomplete, so the forecasting history excludes that partial month from the final training series. $K=4$ is selected for the required four-profile business view even though the global silhouette maximum is $K=2$; this trade-off is explicitly documented. The current held-out churn AUC is 0.829.

Generated from the canonical uploaded Online Retail II workbook. Re-run `run_pipeline.py` to regenerate all numerical artifacts.